Section 5.6 — The Dollar and Risk Appetite

The Plumbing of Liquidity: How Funding Conditions Transmit to Credit Spreads

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Contents

1 Introduction

Sections 5.3–5.5 established a funding-to-credit transmission architecture driven primarily by the reserve quantity channel, with the SOFR–EFFR price channel exhibiting significance only in the 2003–2022 sample (via the TED spread) or regime-conditionally during calm periods. A natural follow-up question is whether this architecture is complete—or whether a global variable mediates the relationship between US funding conditions and credit spreads.

Section 4.5 of this thesis identified the US dollar as the central node in global funding markets: dollar appreciation tightens financial conditions worldwide by raising the cost of dollar-denominated liabilities, constraining the balance-sheet capacity of global financial intermediaries, and compressing their willingness to bear credit risk. This is the global dollar funding channel documented by Bruno and Shin (2015) and Avdjiev et al. (2019). If the dollar is indeed a transmission mechanism, it should satisfy three conditions: (i) it should Granger-cause credit spread changes, (ii) it should respond to—or be linked with—domestic funding variables, and (iii) it should dominate those funding variables in forecast error variance decompositions.

This section tests all three conditions. The analysis proceeds in three parts:

Part A (The Dollar Channel) places the DXY Dollar Index at the centre of the funding-credit system. Using daily data throughout—avoiding the frequency mismatch that complicates the weekly reserve and TGA variables in Sections 5.3–5.4—it estimates two VARs: one embedding DXY within the full funding-credit system (VAR G), and one linking DXY to traded ETF credit ratios (VAR H).

Part B (ETF Robustness) briefly confirms that the Moody's-based findings from earlier sections are robust to ETF-based credit measures.

Part C (Cross-Currency Basis) examines the JPY/USD 3-month cross-currency basis as a proxy for global dollar funding conditions, testing whether it adds predictive content beyond DXY itself.

All variables in this section are observed at daily frequency, which is a methodological improvement over Sections 5.3–5.4 where weekly H.4.1 variables (reserves, TGA) were forward-filled into daily data. Section 5.7 discusses this frequency mismatch limitation in detail.

2 Part A: The Dollar Channel

2.1 Contemporaneous Correlations

Table 1: Contemporaneous Correlations: DXY Returns vs Credit and Funding Variables (Post-2018)

Variable	r	p-value	
ΔBaa (Moody's)	0.111	<0.001	***
ΔAaa (Moody's)	0.103	<0.001	***
Δ(Baa$-$Aaa)	-0.017	0.450	
HYG/SHV return	-0.384	<0.001	***
HYG/LQD return	-0.097	<0.001	***
SOFR$-$EFFR	-0.001	0.951	

ΔReserves	0.053	0.018	**

DXY returns = 100 x d(log DXY). Sample: Apr 2018 – Apr 2026, approx. 1,987 obs.

The contemporaneous correlations reveal two distinct patterns. Dollar appreciation is positively correlated with Moody's spread widening ($r = 0.11$ for ΔBaa, $r = 0.10$ for ΔAaa)—when the dollar strengthens, credit conditions tighten. The same relationship appears with opposite sign in the ETF ratios: $r = -0.38$ for HYG/SHV (a dollar rally compresses the high-yield-to-cash ratio, i.e., widens the implied credit spread). The HYG/SHV correlation is the strongest contemporaneous link found in this thesis.

Critically, DXY returns are uncorrelated with SOFR–EFFR ($r = -0.001$). The dollar channel operates independently of the overnight funding price—it is a risk appetite channel, not a funding cost channel.

2.2 Granger Causality

2.2.1 DXY Predicts Credit Spreads

Table 2: Granger Causality: DXY Returns to Credit Spread Changes (Post-2018)

Direction	Lag	N	F-stat	p-value	
DXY $\rightarrow$ d_Baa	1	1987	24.304	0.0000	***
DXY $\rightarrow$ d_Baa	5	1987	12.746	0.0000	***
DXY $\rightarrow$ d_Baa	10	1987	6.875	0.0000	***
DXY $\rightarrow$ d_Aaa	1	1987	9.889	0.0017	***
DXY $\rightarrow$ d_Aaa	5	1987	6.980	0.0000	***
DXY $\rightarrow$ d_Aaa	10	1987	5.334	0.0000	***
DXY $\rightarrow$ d_Baa_Aaa	1	1987	5.577	0.0183	**
DXY $\rightarrow$ d_Baa_Aaa	5	1987	9.078	0.0000	***
DXY $\rightarrow$ d_Baa_Aaa	10	1987	7.610	0.0000	***
DXY $\rightarrow$ r_HYG_SHV	1	1985	0.323	0.5696	NA
DXY $\rightarrow$ r_HYG_SHV	5	1985	5.238	0.0001	***
DXY $\rightarrow$ r_HYG_SHV	10	1985	1.674	0.0812	*
DXY $\rightarrow$ r_HYG_LQD	1	1985	3.371	0.0665	*
DXY $\rightarrow$ r_HYG_LQD	5	1985	3.307	0.0056	***
DXY $\rightarrow$ r_HYG_LQD	10	1985	1.703	0.0746	*

DXY returns Granger-cause all three Moody's credit variables at every lag tested: ΔBaa ($F = 24.3$, $p < 0.0001$ at lag 1), ΔAaa ($F = 9.9$, $p = 0.002$ at lag 1), and Δ(Baa−Aaa) ($F = 5.6$, $p = 0.018$ at lag 1). All nine tests are significant at the 5% level, with eight at 1%. This is the most uniformly significant Granger result in the thesis—no other variable predicts credit spreads this consistently across both levels and lags.

For ETF credit ratios, DXY → HYG/SHV is significant at lag 5 ($F = 5.24$, $p < 0.001$) and DXY → HYG/LQD at lag 5 ($F = 3.31$, $p = 0.006$). The lag-1 tests for ETFs are weaker, suggesting that the dollar's effect on traded credit prices operates over the weekly horizon.

2.2.2 Bidirectionality: Credit Also Predicts DXY

Table 3: Reverse Granger Causality: Credit Variables to DXY Returns (Post-2018)

Direction	Lag	N	F-stat	p-value	
d_Baa $\rightarrow$ DXY	1	1987	2.524	0.1123	NA
d_Baa $\rightarrow$ DXY	5	1987	2.807	0.0156	**
d_Baa $\rightarrow$ DXY	10	1987	1.948	0.0353	**
d_Aaa $\rightarrow$ DXY	1	1987	1.853	0.1736	NA
d_Aaa $\rightarrow$ DXY	5	1987	5.684	0.0000	***
d_Aaa $\rightarrow$ DXY	10	1987	3.634	0.0001	***
r_HYG_SHV $\rightarrow$ DXY	1	1985	118.182	0.0000	***
r_HYG_SHV $\rightarrow$ DXY	5	1985	25.442	0.0000	***
r_HYG_SHV $\rightarrow$ DXY	10	1985	16.213	0.0000	***
r_HYG_LQD $\rightarrow$ DXY	1	1985	0.030	0.8620	NA
r_HYG_LQD $\rightarrow$ DXY	5	1985	4.197	0.0008	***
r_HYG_LQD $\rightarrow$ DXY	10	1985	2.988	0.0010	***

The dollar-credit link is bidirectional. The strongest reverse channel is HYG/SHV $\rightarrow$ DXY ($F = 118.2$ at lag 1)—when high-yield ETFs fall relative to cash, the dollar strengthens the next day. This is consistent with the flight-to-safety mechanism: credit stress triggers dollar appreciation as investors de-risk into dollar-denominated safe assets. The Moody's-based reverse channel is also significant but weaker (ΔAaa $\rightarrow$ DXY: $F = 5.7$ at lag 5), reflecting the slower updating of model-based spreads compared to ETF prices.

The bidirectionality does not invalidate the DXY $\rightarrow$ credit channel; rather, it reveals a feedback loop. The VAR framework in Section 2.3 below jointly models both directions.

2.2.3 Funding Variables and the Dollar

Table 4: Granger Causality: Funding Variables to DXY (Post-2018)

Direction	Lag	N	F-stat	p-value	
d_Reserves $\rightarrow$ DXY	1	1987	8.754	0.0031	***
d_Reserves $\rightarrow$ DXY	5	1987	2.251	0.0470	**
d_Reserves $\rightarrow$ DXY	10	1987	2.050	0.0254	**
d_TGA $\rightarrow$ DXY	1	1987	0.583	0.4453	NA
d_TGA $\rightarrow$ DXY	5	1987	1.295	0.2631	NA
d_TGA $\rightarrow$ DXY	10	1987	0.944	0.4910	NA
SOFR_EFFR $\rightarrow$ DXY	1	1987	0.000	0.9996	NA
SOFR_EFFR $\rightarrow$ DXY	5	1987	0.263	0.9335	NA
SOFR_EFFR $\rightarrow$ DXY	10	1987	0.347	0.9680	NA

ΔReserves Granger-causes DXY returns at all three lags ($F = 8.75$, $p = 0.003$ at lag 1). Neither TGA nor SOFR–EFFR predicts the dollar. This result connects the reserve quantity channel of Section 5.4 to the dollar channel: when the Fed drains reserves (e.g., through quantitative tightening), the resulting tightening of dollar liquidity propagates to the dollar's exchange value, which in turn tightens credit conditions globally.

In the reverse direction, DXY Granger-causes ΔTGA at lag 1 ($F = 6.05$, $p = 0.014$), likely reflecting the mechanical link between dollar strength and Treasury cash management.

2.3 VAR G: DXY in the Funding-Credit System

We embed DXY in the core funding-credit VAR of Section 5.4, replacing the conventional five-variable ordering with:

$$\text{DXY return} \rightarrow \Delta\text{TGA} \rightarrow \Delta\text{Reserves} \rightarrow \text{SOFR–EFFR} \rightarrow \Delta\text{Baa}$$

The Cholesky ordering places DXY first as the most exogenous variable—a global price that is unlikely to be driven contemporaneously by domestic US funding flows. The remaining variables follow the fiscal-to-monetary-to-credit ordering used in Sections 5.3–5.4.

Table 5: VAR G Specification and Diagnostics

Variables	\$r_{\text{DXY}}\$, \$\Delta\$TGA, \$\Delta\$Res., SOFR–EFFR, \$\Delta\$Baa
Sample	Apr 2018 – Apr 2026
Observations	1,987
Lag order (AIC)	10
Max eigenvalue	0.921 (stable)
Portmanteau (20 lags)	\$\chi^2 = 326.6\$, \$p = 0.001\$
Joint Granger: DXY	\$F = 2.69\$, \$p < 0.0001\$***
Joint Granger: Reserves	\$F = 2.60\$, \$p < 0.0001\$***

The VAR is stable with all eigenvalues inside the unit circle. DXY and ΔReserves are the only two variables that are jointly significant in the multivariate Granger test; ΔTGA ($p = 0.187$) and SOFR–EFFR ($p = 0.930$) are not. This confirms the bivariate finding: the dollar and reserves are the operative channels, while the overnight funding price and fiscal flows do not add predictive content once DXY is included.

The adjusted R^2 for ΔBaa is 8.2%, comparable to the Moody's VARs of Section 5.4. SOFR–EFFR achieves $R^2 = 19.6\%$ (its own persistence, documented in Section 5.2). DXY's own R^2 is 1.7%—consistent with efficient-market expectations for a major currency return.

2.3.1 Impulse Response Functions

The IRF panel reveals a sharp, statistically significant response: a one-standard-deviation DXY shock widens ΔBaa on impact and through approximately five days, with the confidence interval excluding zero for the first three horizons. This is a faster and more significant response than any funding variable achieves in the VARs of Section 5.4.

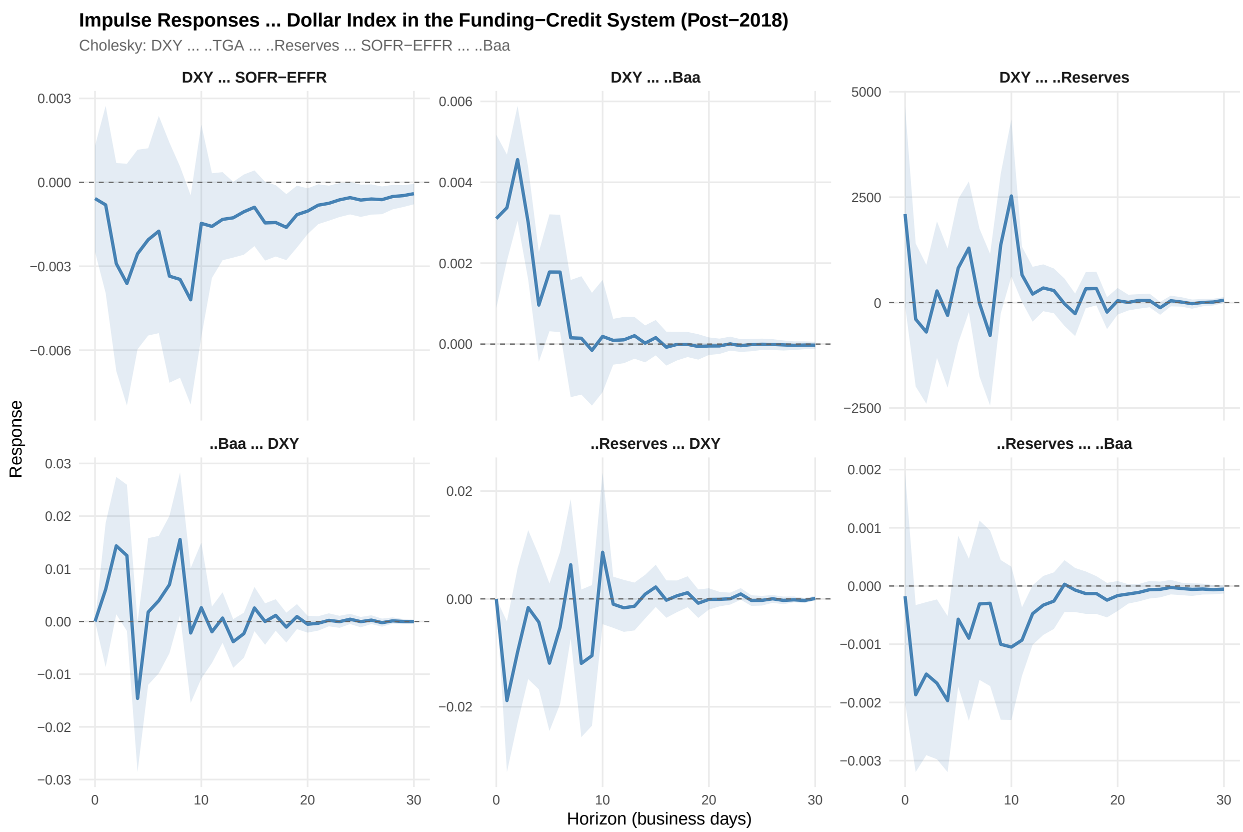

Figure 1: Impulse Response Functions — VAR G: Dollar Index in the Funding-Credit System (Post-2018). Cholesky ordering: DXY → ΔTGA → ΔReserves → SOFR-EFFR → ΔBaa. Shaded: 95% bootstrap CI (500 replications).

The reverse channel (ΔBaa $\rightarrow$ DXY) shows a positive response—credit spread widening leads to dollar appreciation—consistent with the flight-to-safety interpretation. The ΔReserves $\rightarrow$ DXY response is initially negative, suggesting that reserve injections depreciate the dollar, consistent with the liquidity-provision narrative.

2.3.2 Forecast Error Variance Decomposition

Table 6: FEVD of ΔBaa: Dollar Shocks Dominate All Funding Variables (VAR G, Post-2018)

h	DXY (\%)	ΔTGA (\%)	ΔRes. (\%)	SOFR (\%)	Own (\%)
1	1.0	0.0	0.0	0.0	99.0
5	4.9	0.2	1.2	0.1	93.6
10	5.4	0.3	1.4	0.1	92.7
20	5.4	0.3	1.6	0.2	92.5
30	5.4	0.3	1.6	0.2	92.4

Total funding contribution at h=30: 7.5 pct (DXY: 5.4, Reserves: 1.6, TGA: 0.3, SOFR: 0.2).

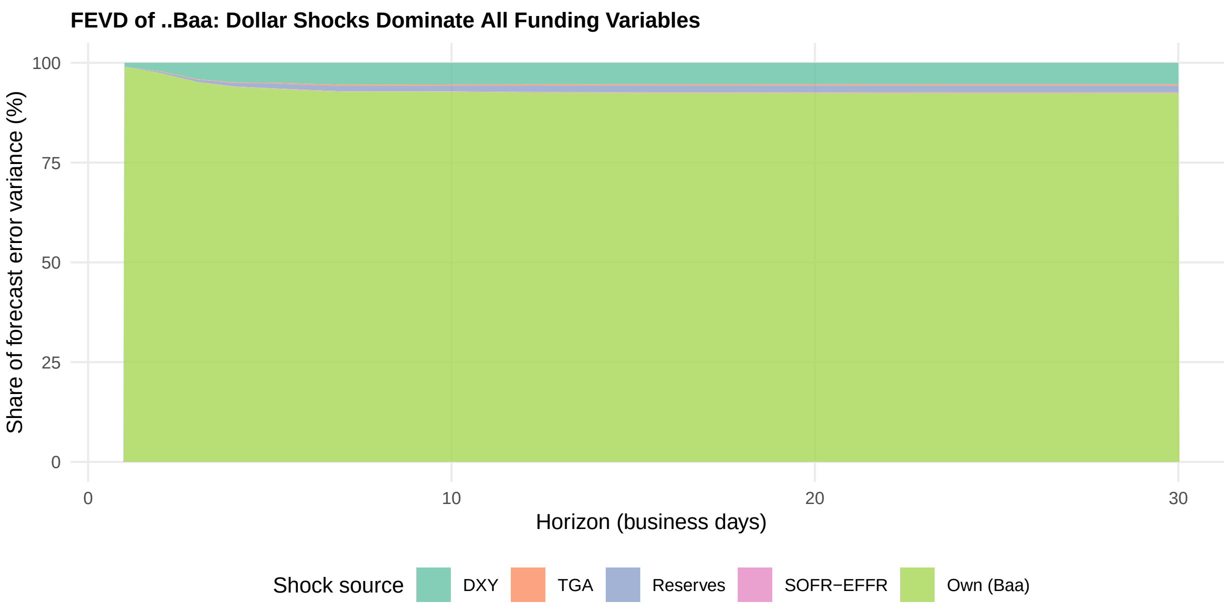

Figure 2: FEVD of ΔBaa: Stacked area decomposition showing DXY as the dominant external shock. At $h = 30$, DXY explains 5.4% of Baa forecast error variance—more than all domestic funding variables combined (2.1%).

This is the central quantitative result of the section. At the 30-day horizon, DXY shocks explain **5.4%** of ΔBaa forecast error variance. The domestic funding variables explain a combined **2.1%** (Reserves: 1.6%, TGA: 0.3%, SOFR–EFFR: 0.2%). The dollar alone accounts for more than twice the variance contribution of all three domestic funding channels.

The DXY variable itself is extremely exogenous: 97.4% of its own forecast error variance at $h = 30$ is due to own shocks, confirming that the Cholesky ordering (DXY first) is appropriate and that the dollar is not being driven by the other variables in the system.

2.4 VAR H: DXY and ETF Credit Ratios

To test whether the dollar channel transmits to traded credit prices, we estimate a second VAR:

$$r_{\text{DXY}} \rightarrow \Delta\text{Reserves} \rightarrow r_{\text{HYG/SHV}} \rightarrow r_{\text{HYG/LQD}}$$

where HYG/SHV captures the credit-to-cash spread and HYG/LQD captures the quality differential, both using daily ETF returns.

Table 7: VAR H Specification: DXY and ETF Credit Ratios

Observations	1,985
Lag order (AIC)	15
Max eigenvalue	0.914 (stable)
R^2 HYG/SHV	12.6\%
R^2 HYG/LQD	13.2\%

2.4.1 ETF FEVD Results

The FEVD reveals a striking contrast between the two credit measures. DXY explains **14.8%** of HYG/SHV variance on impact and **13.9%** at $h = 30$—a contemporaneous and persistent effect. For HYG/LQD (the quality differential), DXY's contribution is much smaller (0.7% at $h = 1$, 2.2% at $h = 30$), while the cross-credit channel from HYG/SHV dominates (17.1% at $h = 30$).

The interpretation is clear: the dollar channel primarily affects the overall credit risk premium (HYG/SHV), not the quality differential (HYG/LQD). Dollar appreciation compresses risk appetite uniformly rather than differentially widening the gap between high-yield and investment-grade.

2.5 Interpretation: The Dollar as Missing Link

The results of Part A establish the dollar as the strongest single predictor of credit spread dynamics in this thesis. Three observations frame the interpretation:

1. The dollar dominates domestic funding variables. In the FEVD of ΔBaa, DXY alone (5.4%) explains more than ΔReserves (1.6%), ΔTGA (0.3%), and SOFR–EFFR (0.2%) combined. This does not invalidate the funding channels of Sections 5.3–5.5—those channels operate, and ΔReserves remains jointly significant in VAR G—but it places them in a broader context. The reserve channel may partly operate *through* the dollar: reserve drains tighten dollar liquidity (ΔReserves → DXY is significant), which then transmits to credit.

2. The channel is a risk appetite channel, not a funding cost channel. DXY is uncorrelated with SOFR–EFFR ($r = -0.001$) and SOFR–EFFR does not Granger-cause DXY. The dollar's effect on credit operates through global risk appetite and balance-sheet capacity, not through the overnight funding rate. This is consistent with the Bruno–Shin (2015) mechanism: dollar appreciation raises the effective leverage of global intermediaries with currency mismatches, reducing their risk-bearing capacity.

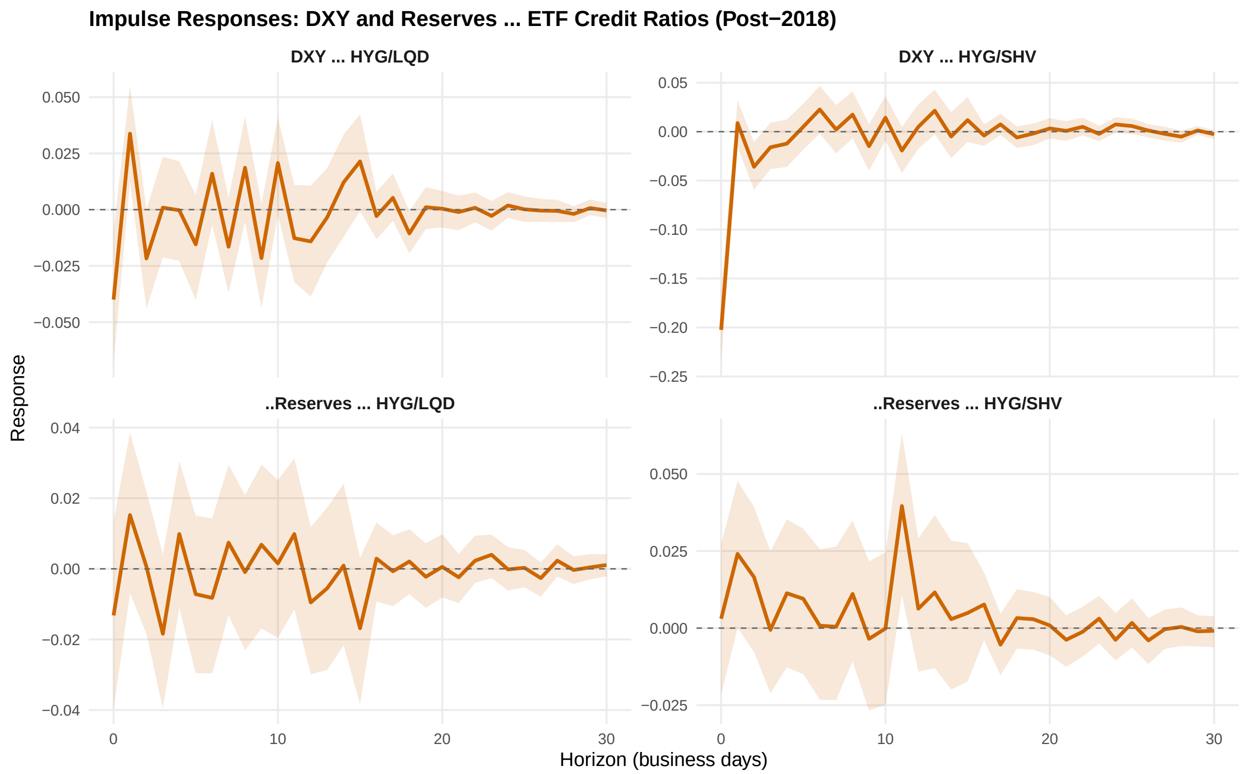

Figure 3: Impulse Response Functions: DXY and Reserves to ETF Credit Ratios (VAR H, Post-2018). DXY shocks produce a sharp, significant response in HYG/SHV (credit-to-cash). Shaded: 95% bootstrap CI.

3. The bidirectional feedback loop is consistent with theory. The Brunnermeier and Pedersen (2009) margin spiral predicts that asset price declines tighten funding conditions, which further depress asset prices. In the dollar-credit system, this manifests as: credit stress → flight to dollar → dollar appreciation → tighter global funding → further credit widening. The bivariate Granger tests confirm both legs of this loop.

3 Part B: ETF Robustness

This part briefly confirms that the funding channels documented with Moody's spreads in Sections 5.3–5.4 are robust when measured with traded ETF prices.

Table 8: Significant Granger Tests: Funding Variables to ETF Credit Measures (Post-2018)

Direction	Lag	N	F-stat	p-value	
d_Reserves $\rightarrow$ r_HYG	1	2002	11.371	0.0008	***
d_Reserves $\rightarrow$ r_HYG	5	2002	2.466	0.0309	**
d_Reserves $\rightarrow$ d_log_HYG_IEF	1	2002	6.110	0.0135	**
d_Reserves $\rightarrow$ d_log_HYG_IEF	5	2002	2.342	0.0394	**
d_Reserves $\rightarrow$ d_log_LQD_IEF	5	2002	2.435	0.0328	**
d_TGA $\rightarrow$ r_HYG	1	2002	7.022	0.0081	***
d_TGA $\rightarrow$ d_log_HYG_IEF	1	2002	7.836	0.0052	***

d__TGA $\rightarrow$ d__log__HYG__IEF	5	2002	2.043	0.0698	*
d__TGA $\rightarrow$ d__log__LQD__IEF	1	2002	7.567	0.0060	***

9 of 24 tests significant at 10 pct level. *** p<0.01, ** p<0.05, * p<0.10.

Of 24 bivariate Granger tests (3 funding variables × 4 ETF measures × 2 lags), 9 are significant at the 10% level. The pattern matches the Moody's results:

- **Reserve channel confirmed.** ΔReserves Granger-causes HYG returns ($F = 11.37$, $p < 0.001$ at lag 1) and both duration-hedged ratios HYG/IEF and LQD/IEF at the weekly lag.
- **TGA channel strengthened.** ΔTGA Granger-causes HYG ($F = 7.02$, $p = 0.008$), HYG/IEF ($F = 7.84$, $p = 0.005$), and LQD/IEF ($F = 7.57$, $p = 0.006$) all at lag 1. The TGA effect is stronger in the ETF framework than in the Moody's analysis, possibly reflecting faster price discovery in traded markets.
- **SOFR–EFFR remains insignificant.** No SOFR → ETF test reaches significance, confirming the post-2018 price channel null.

4 Part C: Cross-Currency Basis

4.1 Context and Data

The 3-month JPY/USD cross-currency basis measures the premium for borrowing US dollars through FX swap markets relative to direct USD borrowing. A more negative basis signals a larger dollar funding premium—higher demand for dollar liquidity from non-US institutions. Our data covers May 2024–May 2026 (495 observations), with the basis averaging −31.7 bp ($\sigma = 5.5$ bp).

4.2 Correlations and Granger Causality

The level of the JPY basis exhibits strong correlations: $r = -0.65$ with DXY and $r = 0.66$ with Aaa spreads. In first differences, Δ(JPY basis) correlates with ΔBaa ($r = -0.28$) and with DXY returns ($r = -0.13$): when the dollar strengthens, the basis becomes more negative (dollar funding tightens), consistent with the global dollar funding channel.

Table 9: Granger Causality: Cross-Currency Basis Sample (May 2024 – May 2026)

Direction	Lag	N	F-stat	p-value	
SOFR__EFFR $\rightarrow$ d__JPY__basis	1	493	1.342	0.2472	NA
SOFR__EFFR $\rightarrow$ d__JPY__basis	5	493	0.826	0.5313	NA
d__Reserves $\rightarrow$ d__JPY__basis	1	494	0.086	0.7692	NA
d__Reserves $\rightarrow$ d__JPY__basis	5	494	0.487	0.7861	NA
r__DXY $\rightarrow$ d__JPY__basis	1	489	3.883	0.0493	**
r__DXY $\rightarrow$ d__JPY__basis	5	489	4.376	0.0007	***
d__JPY__basis $\rightarrow$ d__Baa	1	492	0.001	0.9719	NA
d__JPY__basis $\rightarrow$ d__Baa	5	492	0.333	0.8930	NA

d_JPY_basis $\rightarrow$ d_Aaa	1	492	0.130	0.7186	NA
d_JPY_basis $\rightarrow$ d_Aaa	5	492	0.135	0.9843	NA
d_JPY_basis $\rightarrow$ d_Baa_Aaa	1	492	0.402	0.5262	NA
d_JPY_basis $\rightarrow$ d_Baa_Aaa	5	492	1.320	0.2543	NA
d_Baa $\rightarrow$ d_JPY_basis	1	492	9.957	0.0017	***
d_Baa $\rightarrow$ d_JPY_basis	5	492	5.186	0.0001	***
d_Aaa $\rightarrow$ d_JPY_basis	1	492	15.257	0.0001	***
d_Aaa $\rightarrow$ d_JPY_basis	5	492	7.235	0.0000	***
d_JPY_basis $\rightarrow$ r_DXY	1	489	0.330	0.5657	NA
d_JPY_basis $\rightarrow$ r_DXY	5	489	1.275	0.2734	NA

The Granger results establish a clear causal hierarchy:

DXY and credit predict the basis. DXY Granger-causes Δ(JPY basis) at both lags ($F = 4.38$, $p < 0.001$ at lag 5). ΔAaa and ΔBaa also strongly predict basis changes ($F = 15.26$ and $F = 9.96$ at lag 1). Dollar appreciation and credit spread widening lead to further deterioration of the cross-currency basis.

The basis does not predict credit or the dollar. Δ(JPY basis) fails to Granger-cause ΔBaa, ΔAaa, or Δ(Baa−Aaa) at any lag. Nor does it predict DXY returns. The basis is a *symptom* of dollar funding stress, not a *cause*—at least over this sample.

No domestic funding-to-basis channel. Neither SOFR–EFFR, ΔReserves, nor ΔTGA predicts the basis. Domestic overnight funding dynamics do not drive the cross-currency basis, which reflects global institutional demand for dollar hedging.

4.3 Rolling Correlations

The 60-day rolling correlation between DXY returns and basis changes fluctuates between -0.4 and $+0.2$, confirming the contemporaneous relationship is time-varying. Periods of sustained negative correlation coincide with episodes of dollar strengthening and basis widening.

5 Summary

This section yields four findings that reshape the narrative of the thesis:

1. **The dollar is the strongest predictor of credit spreads.** DXY Granger-causes ΔBaa, ΔAaa, and Δ(Baa−Aaa) at every lag tested—the most uniformly significant result in this chapter. In the FEVD, DXY alone (5.4%) explains more than all domestic funding variables combined (2.1%).

2. **The dollar channel operates through risk appetite, not funding costs.** DXY is uncorrelated with SOFR–EFFR and operates independently of the overnight funding price. The mechanism is balance-sheet-based: dollar appreciation tightens the effective leverage of global intermediaries, reducing their willingness to bear credit risk (Bruno and Shin, 2015).

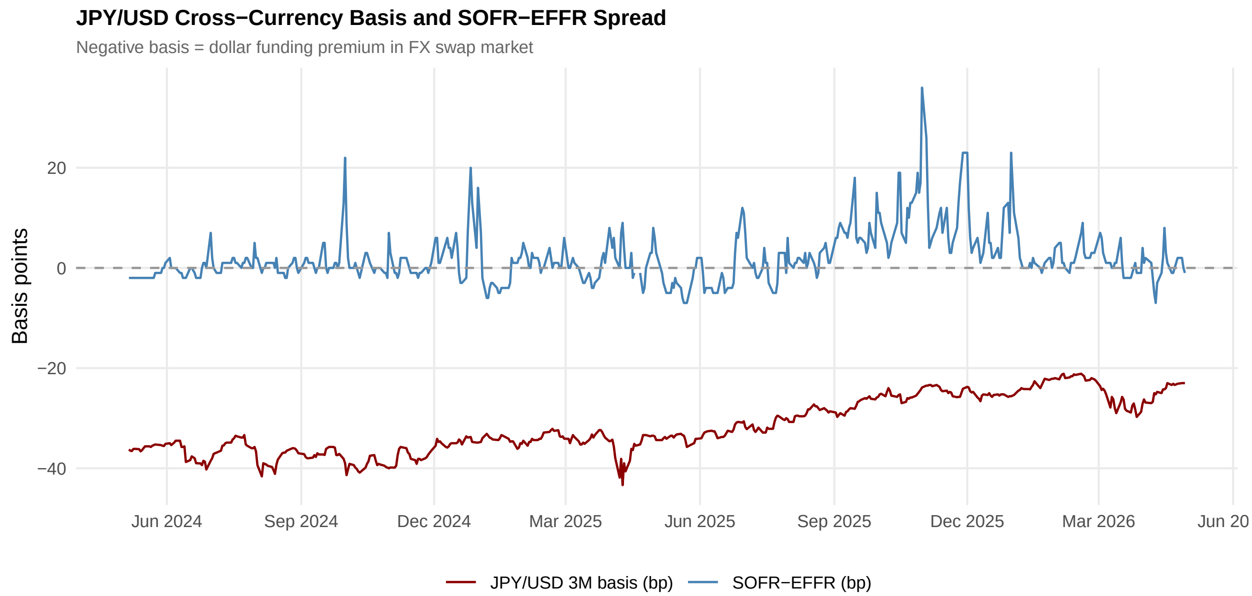

Figure 4: JPY/USD 3-month cross-currency basis and SOFR–EFFR spread. More negative basis = higher dollar funding premium. Sample: May 2024 – May 2026.

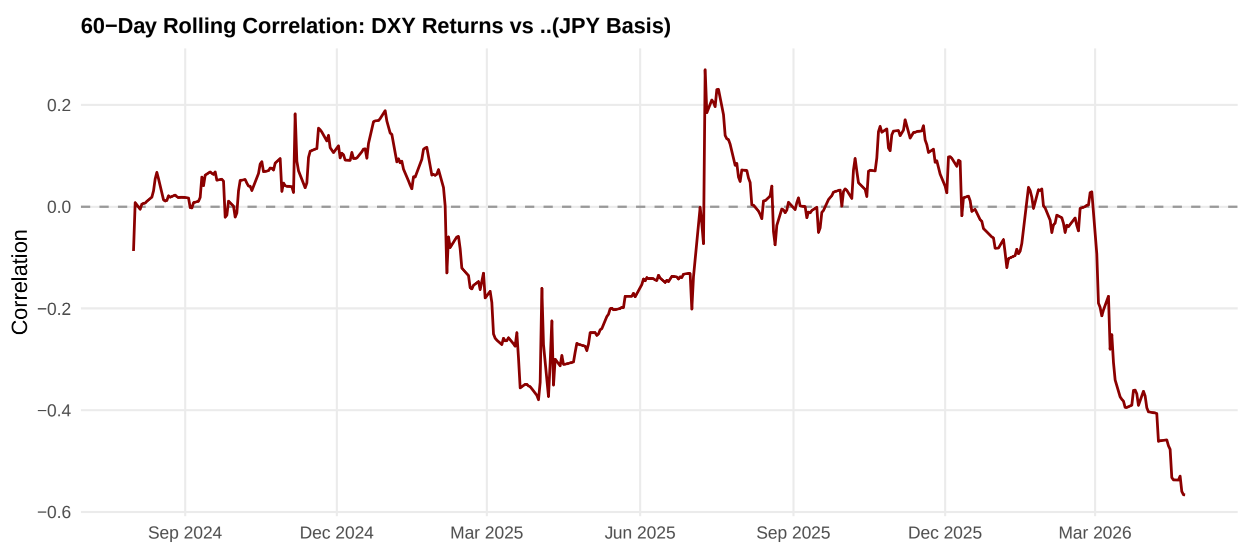

Figure 5: 60-day rolling correlation: DXY returns vs Δ(JPY basis). The relationship is time-varying but predominantly negative—dollar appreciation coincides with basis deterioration.

3. **The reserve channel partly operates through the dollar.** ΔReserves Granger-causes DXY at lag 1 ($F = 8.75$, $p = 0.003$), while both remain independently significant in VAR G. This suggests a transmission chain: reserve drains $\rightarrow$ dollar appreciation $\rightarrow$ credit widening, with the dollar serving as an intermediate link.

4. **The cross-currency basis is a symptom, not a cause.** The JPY/USD basis does not predict credit spreads or DXY returns over the available 2-year sample. Instead, DXY and credit changes lead the basis. This is an honest null: the Bruno–Shin global funding channel may activate during crises (2008, 2020) absent from our sample. A longer time series would be needed to test this rigorously.

6 References

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